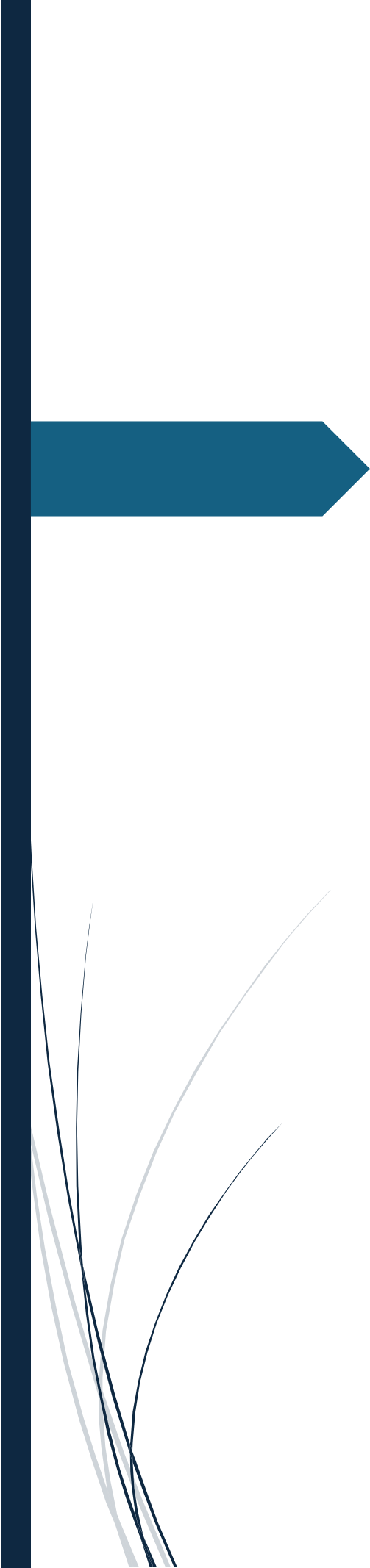
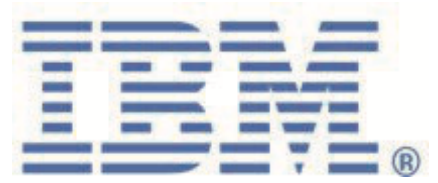


A decorative graphic on the left side of the page. It features a thick, dark blue vertical bar. A horizontal blue arrow points to the right from the middle of this bar. At the bottom of the vertical bar, there are several thin, curved, light blue lines that sweep upwards and to the right.

Maximo Monitor 9.1 Managed Gateway Modbus Lab

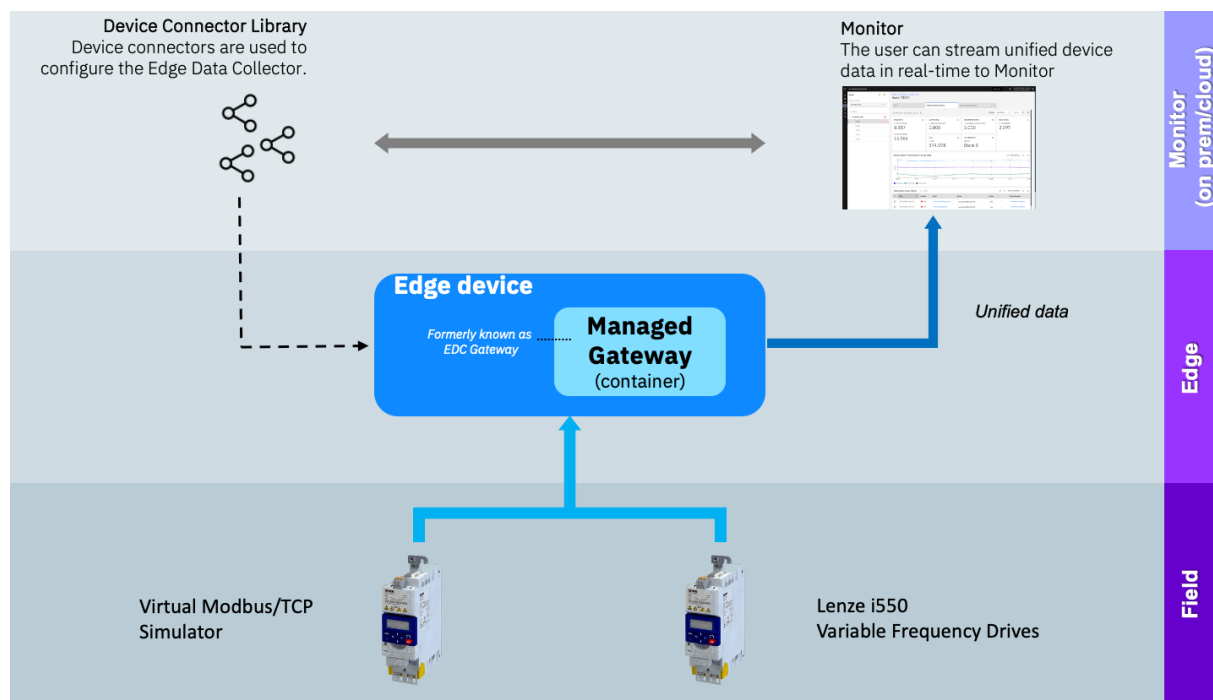
Table of Contents

Overview.....	3
Exercise 1. Setup Simulator Environment	4
Exercise 2. Create a Managed Gateway.....	6
Exercise 3. Add an Industrial Device	8
Exercise 4. Deploy and Verify Data Inflow	11
Exercise 5. Add another Industrial Device.....	15
Exercise 6. Deploy and View in Dashboard	17
End	

Overview

In this lab you will learn how to setup a Managed Gateway within Maximo Monitor 9.1 and adding a couple of OT devices from the Device Library.

You will learn the steps needed to successfully setup a Managed Gateway using some virtual simulated Variable Frequency Drives (VFD) that will deliver the device data over Modbus/TCP via the Managed Gateway into Maximo Monitor using the pre-configured connector for Lenze i550 VFD from the Device Connector Library.



The exercises will cover:

- Setting up the simulator environment
- Create a Managed Gateway
- Add devices to the Manage Gateway
- Verify data flow from Modbus/TCP Variable Frequency Drives all the way into Maximo Monitor

Exercise 1. Setup Simulator Environment

In this Exercise you will learn how to: Install the Modbus simulator

1. You need the IP address of the machine where you run the simulator no matter which version you are using. You will need that later when configuring the Managed Gateway.

- **Get IP Address on macOS**

Use the following command: **ifconfig en0**

```
[tanmayparashar@Tanmays-MacBook-Pro-2 ~ % ifconfig en0
en0: flags=8863<UP, BROADCAST, SMART, RUNNING, SIMPLEX, MULTICAST> mtu 1500
    options=6460<TSO4, TSO6, CHANNEL_IO, PARTIAL_CSUM, ZEROINVERT_CSUM>
    ether da:63:8e:d0:95:02
    inet6 fe80::1c86:af08:7369:24a1%en0 prefixlen 64 secured scopeid 0xe
    inet 192.168.1.3 netmask 0xffffff00 broadcast 192.168.1.255
    nd6 options=201<PERFORMNUD, DAD>
    media: autoselect
    status: active
```

- **Get IP Address on LINUX**

Use the following command: **hostname -I**

```
+ ~ ssh pi@192.168.1.97
pi@192.168.1.97's password:
Linux RPE007 6.1.21-v7+ #1642 SMP Mon Apr  3 17:20:52 BST 2023 armv7l

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Jun 19 11:22:38 2024 from 192.168.1.42
pi@RPE007:~$ hostname -I
192.168.1.97 fde1:7a3d:7b2a:ef4e:200a:c9c5:dd61:a37e
pi@RPE007:~$
```

- **Get IP Address on Windows**

Use the following command: **ipconfig**

```
Command Prompt
Microsoft Windows [Version 10.0.19045.2364]
(c) Microsoft Corporation. All rights reserved.

C:\Users\014420678>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : fde1:7a3d:7b2a:ef4e:378a:4b4b:3dc7:3393
    Temporary IPv6 Address. . . . . : fde1:7a3d:7b2a:ef4e:20fc:a178:fb5f:022a
    Link-local IPv6 Address . . . . . : fe80::c73a:b3c4:5a19:9883%2
    IPv4 Address. . . . . : 192.168.1.43
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

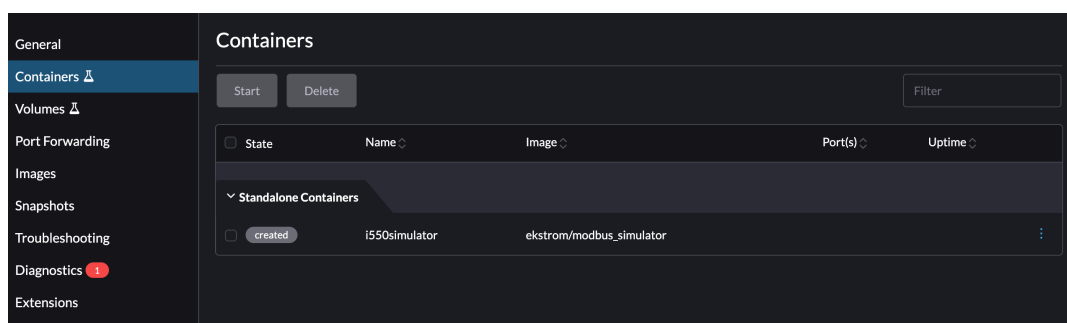
C:\Users\014420678>
```

2. **Install Docker:** There are different ways to install the Docker engine depending of the package and the OS. One multi-platform option is **Rancher Desktop**.
3. It is fairly easy to install Rancher Desktop, as you just have to follow this guide: [Running Docker locally](#)
4. **Create the Docker container:** Open a terminal or command window and run the following command:

docker create -p 10502:10502 -p 20502:20502 -v node_red_data_vol:/data --name i550simulator ekstrom/modbus_simulator

```
tanmayparashar@Tanmays-MacBook-Pro-2 ~ % docker create -p 10502:10502 -p 20502:20502 -v node_red_data_vol:/data --name i550simulator ekstrom/modbus_simulator
8441c31b3dce75686bd79de6c9743031d14c72c39f12ac5fed10231f508dd0b7
tanmayparashar@Tanmays-MacBook-Pro-2 ~ %
```

You can see a container got created in Rancher Desktop

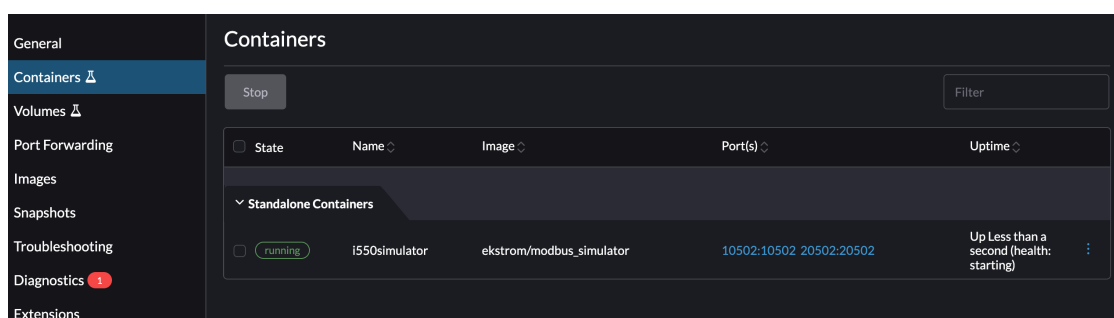


5. **Start the Docker container:** Run the following command to start the container.

docker start i550simulator

```
[tanmayparashar@Tanmays-MacBook-Pro-2 ~ % docker start i550simulator
i550simulator
tanmayparashar@Tanmays-MacBook-Pro-2 ~ %
```

You can see simulator is running in Rancher Desktop



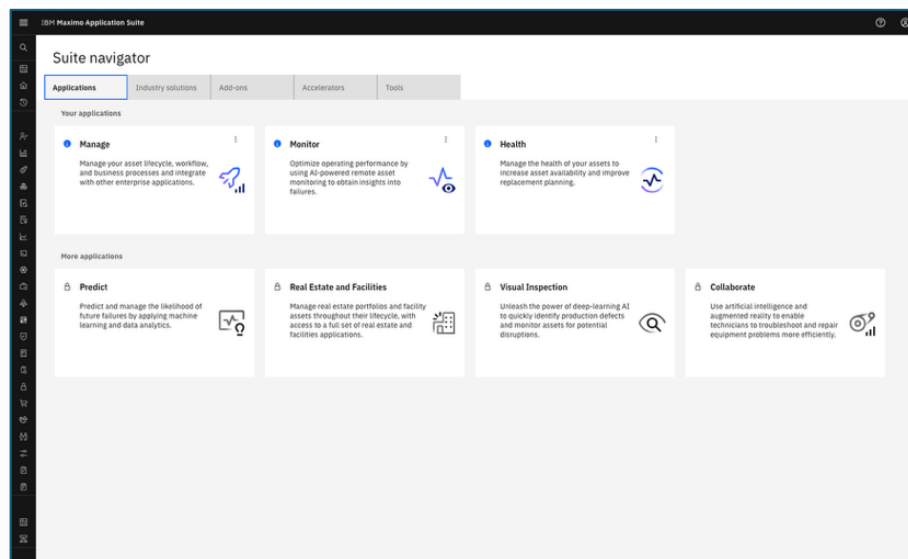


Note: The simulator is now active and the random and dynamic values will change every 30 second. It will run in the background and not produce any output in the terminal/command window.

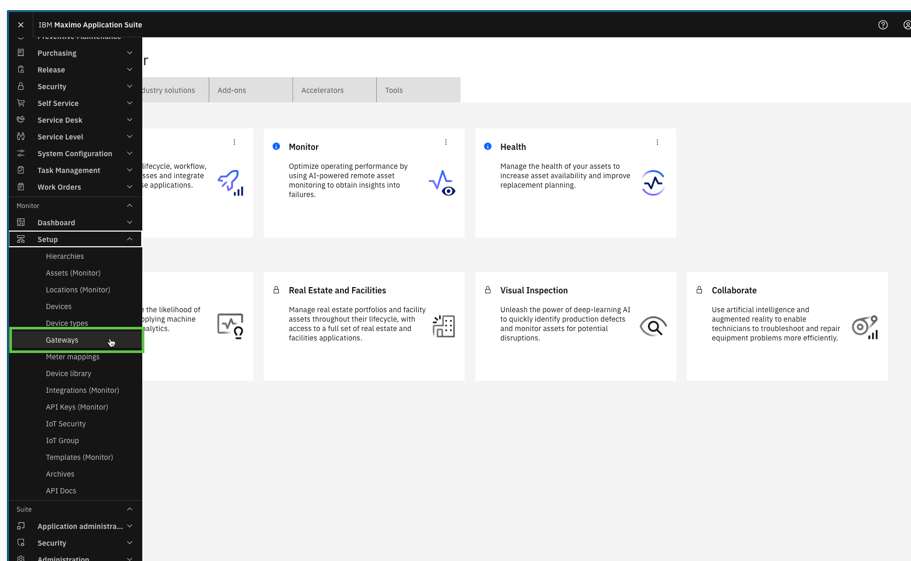
Exercise 2. Create a Managed Gateway

In this Exercise you will learn how to create the Managed Gateway in Monitor.

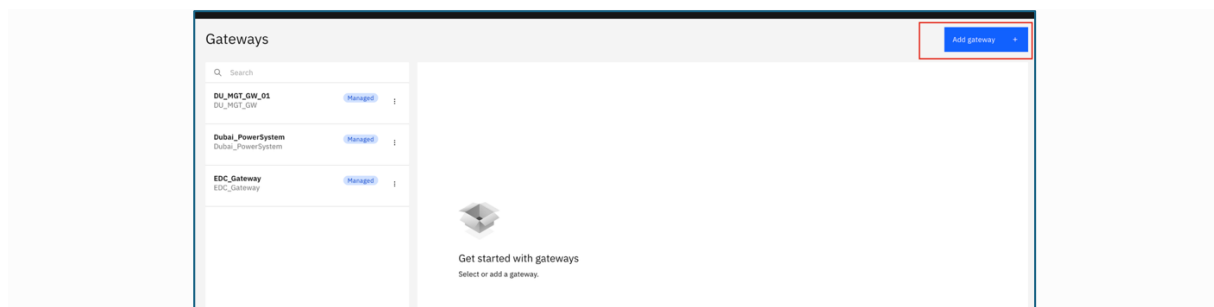
1. Login to Maximo Application Suite with your credentials.



2. Expand Setup under the Monitor section in the left menu and select **Gateways**:

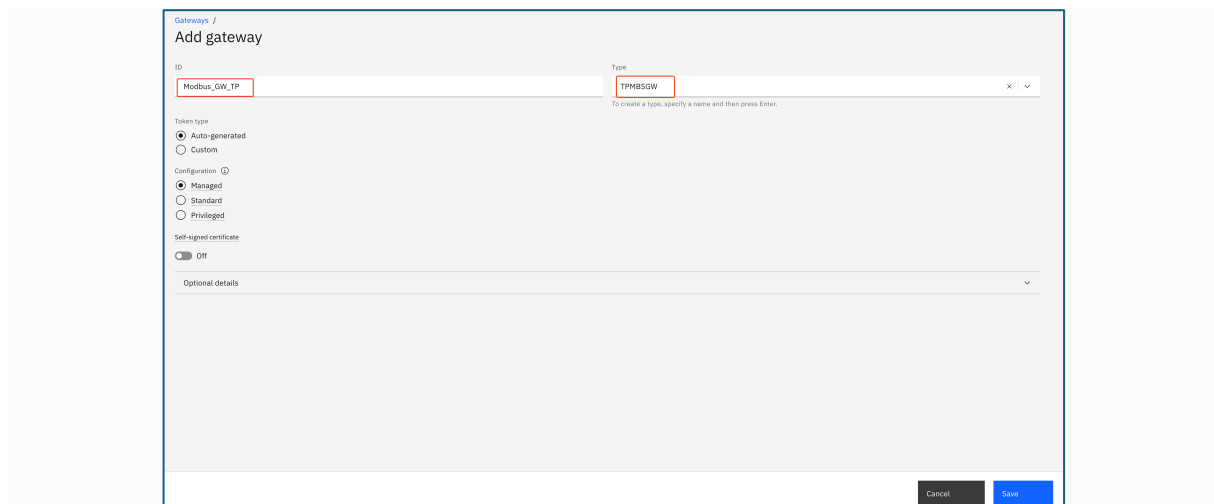


3. Select **Add Gateway**.



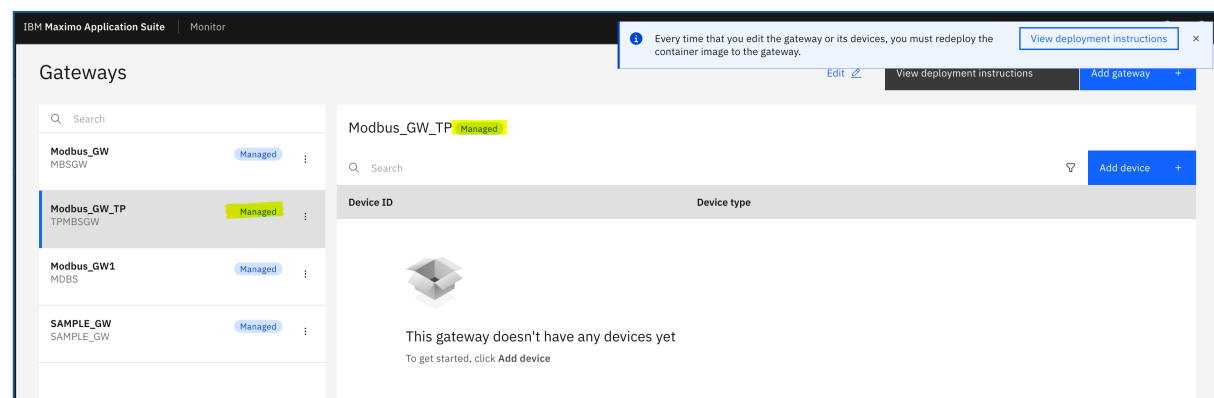
4. Provide gateway id and gateway type name. You can type preferred name and click enter.

- ID- **Modbus_GW_XX** (Replace XX with your initials)
- Type- **XXMBSGW** (Replace XX with your initials)
- Token Type- **Auto-generated**
- Configuration- **Managed**



5. Click on **Save**.

6. You will now see your new Managed Gateway, including a **Managed** tag in both the list of Gateways as well as in the gateway definition.



Exercise 3. Add an Industrial Device

In this Exercise you will learn how to add the first industrial device to the Managed Gateway.

1. Click on **Add Device** and select **Use device library**. (Contd. From last exercise)

Modbus_GW_TP

Add device to TPMBSGW

Configuration
Managed

Managed gateways support only devices from the library.

Use device library ☒ **Add custom device**

Choose a device from a library of preconfigured, manufacturer-specific devices. [Learn more about the library](#)

Configure your own device. Specify custom device details and metrics. [Learn more about custom devices](#)

Cancel Continue

2. Search for **Lenze** in the manufacturer drop-down and select it. Click **Next**:

IBM Maximo Application Suite Monitor

Gateway / Add device

Select device

Select a manufacturer and then search for a device in the library.

Manufacturer

Lenze

Lenze Jantza Johar Kopware Legrand

No devices to show

Cancel Next

3. Select the **i550** product and click on **Next**.

IBM Maximo Application Suite Monitor

Gateway / Add device

Select device

Select a manufacturer and then search for a device in the library.

Manufacturer

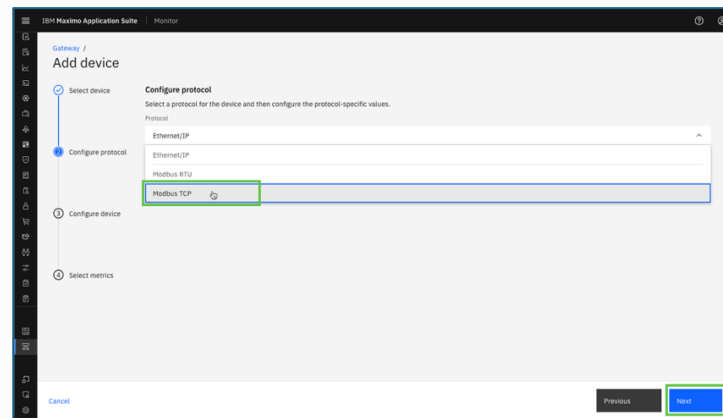
Lenze

Search

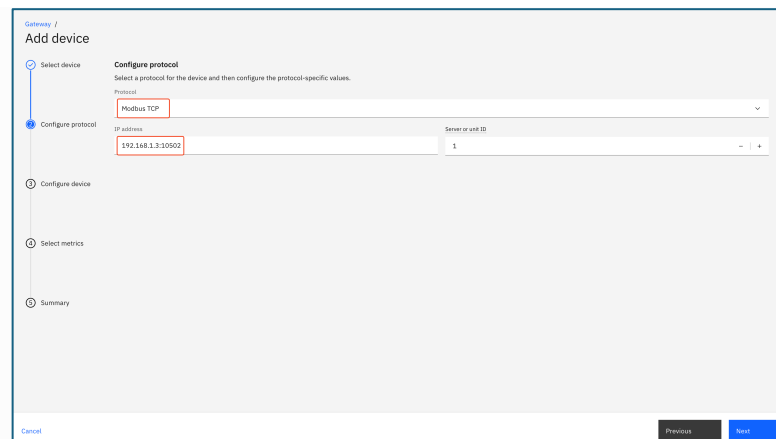
Product name	Product series	Product type
☐ 9400 Highline	E94	Variable Frequency Drives
☐ BaseLine D	8400	Variable Frequency Drives
☐ i530	i500	Variable Frequency Drives
☒ i550	i500	Variable Frequency Drives
☐ i550 protac	i500	Variable Frequency Drives
☐ IP65	SMVector	Variable Frequency Drives
☐ Motec	8400	Variable Frequency Drives
☐ NEHA-1	SHV	Variable Frequency Drives

Cancel Next

4. Select the **Modbus TCP** protocol. Click on **Next**:

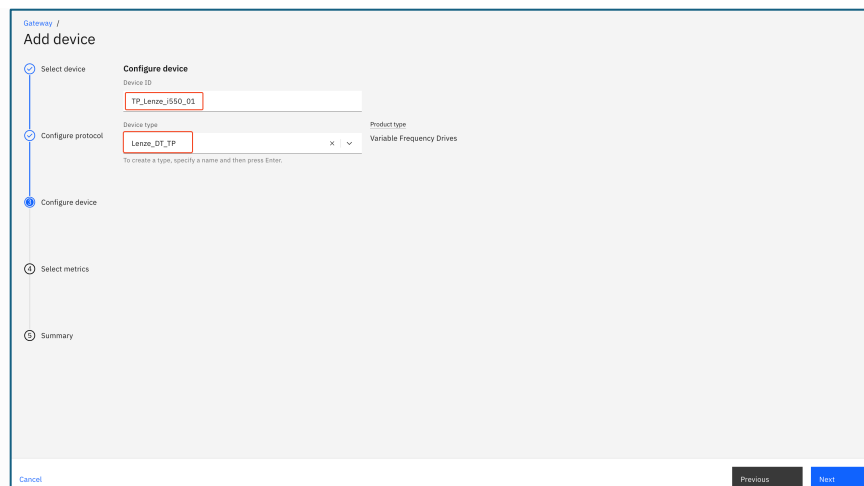


5. Now it is time to use the IP address of the simulator and combine it with port number 10502 separated by : , like **192.168.1.64:10502**. Click on **Next**.



6. It will ask you to configure device and provide your own device id and device type. Click on **Next**.

- Device ID- **XX_Lenze_i550_01**
- Device type- **Lenze_DT_XX** (Replace XX with your initials)



7. Select these three metrics and click on **Next**.

Gateway / Add device

Select device

Configure protocol

Configure device

Select metrics

Summary

Cancel

Previous

Next

Select metrics

Select metrics for the device and configure when data is sent. Selected metrics are added to the device as raw metrics. All of the metrics in the table are added to the selected device type.
[Learn more about metrics](#)

Data frequency (in milliseconds)

1000

Change threshold (%)

On every change

Standard Synthesised

3 items selected

Metric	Unit	Data frequency (ms)	Change threshold (%)
☒ Control Card Temperature	degree Celsius	1000	On every change
☒ DC Bus Voltage	volt	1000	On every change
☒ Frequency Command	hertz	1000	On every change
☐ Motor Current	ampere	1000	On every change
☐ Motor Torque	percent	1000	On every change
☐ Motor Voltage	volt	1000	On every change
☐ Output Frequency	hertz	1000	On every change
☐ Torque Command	percent	1000	On every change
☐ Supply Voltage	volt	1000	On every change

8. Click on **Next** -> **Save**.

Gateway / Add device

Select device

Configure protocol

Configure device

Select metrics

Summary

Cancel

Previous

Save

Summary

You're almost done! Here's a summary of the device configuration details for your confirmation.

Device Details

Manufacturer: Lenze

Product name: i550

Gateway ID: Modbus_GW_TP

Gateway type: TPMBSGW

Device ID: TP_Lenze_i550_01

Device type: Lenze_DT_TP

Protocol Details

Protocol: modbus-tcp

IP/Port/ServerId: 192.168.1.3:10502/1

9. You have now successfully added the first OT device to your Managed Gateway.

IBM Maximo Application Suite | Monitor

Every time that you edit the gateway or its devices, you must redeploy the container image to the gateway. [View deployment instructions](#)

Gateways

Search

Modbus_GW MBSGW Managed

Modbus_GW_TP TPMBSGW Managed

Modbus_GW1 MDBG Managed

SAMPLE_GW SAMPLE_GW Managed

Modbus_GW_TP Managed

Search

Device ID

Device type

TP_Lenze_i550_01 Lenze_DT_TP

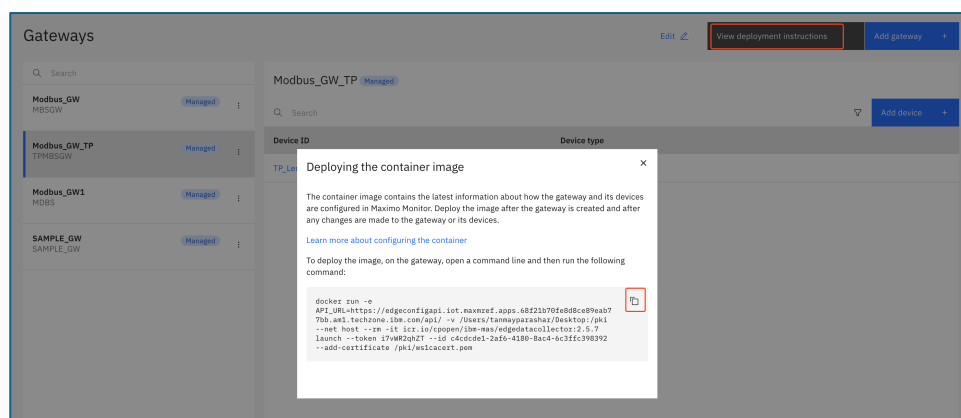
Add device

Exercise 4. Deploy and verify data inflow

In this Exercise you will learn how to:

- Deploy the Managed Gateway
- Verify data inflow

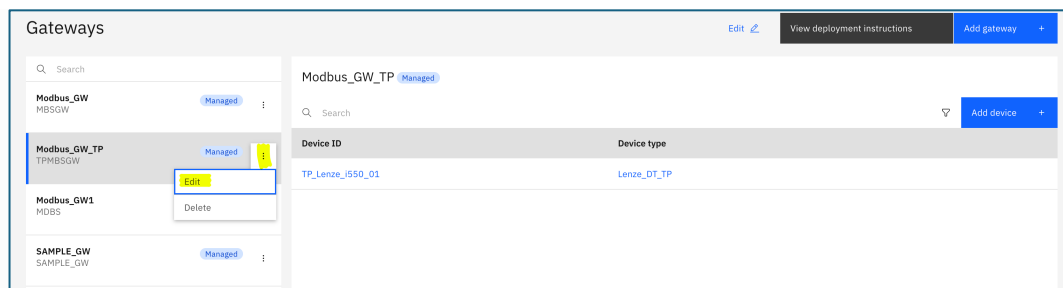
1. While looking at your Managed Gateway in the Gateways list, press the **View deployment instructions**. Click on the docker command to copy it to the clipboard.



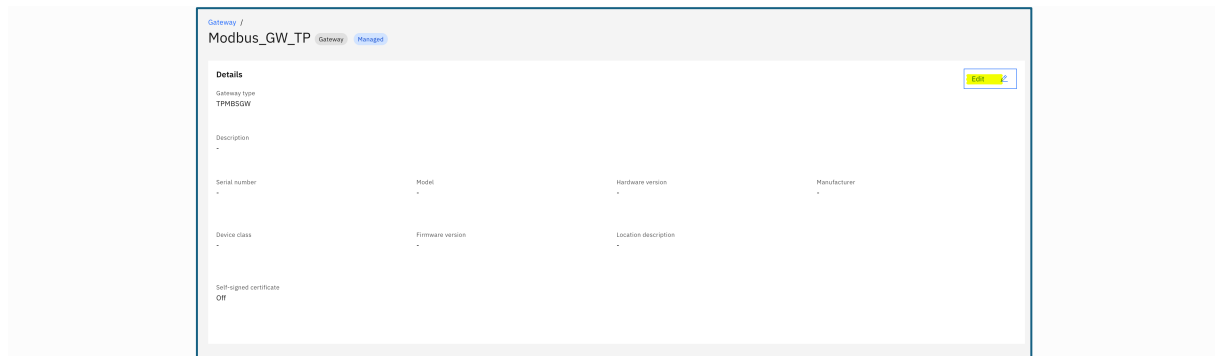
2. Once you run this docker command in Terminal/Command Prompt you will encounter the below error.

```
tanmayparashar@Tanmays-MacBook-Pro-2 ~ % docker run -e API_URL=https://edgeconfigapi.iot.maxmref.apps.68f21b70fe8d8ce89eab77bb.am1.techzone.ibm.com/api/ --net host --rm -it icr.io/cpopen/ibm-mas/edgedatacollector:2.5.7 launch --token i7vWR2qhZT --id c4cdcd1-2af6-4180-8ac4-6c3ffc398392
WARNING: The requested image's platform (linux/amd64) does not match the detected host platform (linux/arm64/v8) and no specific platform was requested
2025-10-26T18:24:07.208824Z INFO: edge-cli/src/lib.rs:186: Arguments: launch --token i7vWR2qhZT --id c4cdcd1-2af6-4180-8ac4-6c3ffc398392
2025-10-26T18:24:07.211473Z INFO: edge-cli/src/lib.rs:188: Version: 2.5.7
2025-10-26T18:24:07.211592Z INFO: edge-cli/src/lib.rs:189: Git sha1: cab6ca6f2bf101743db0f093cd83bb9763bc4668
2025-10-26T18:24:07.211609Z INFO: edge-cli/src/lib.rs:110: Branch: master
2025-10-26T18:24:07.211656Z INFO: edge-cli/src/lib.rs:111: Build number: 748
2025-10-26T18:24:07.211700Z INFO: edge-cli/src/lib.rs:112: Build ID: 41027330
2025-10-26T18:24:07.347811Z INFO: /omniot/edge/edge-rest/src/lib.rs:277: GET https://edgeconfigapi.iot.maxmref.apps.68f21b70fe8d8ce89eab77bb.am1.techzone.ibm.com/api/installations/c4cdcd1-2af6-4180-8ac4-6c3ffc398392/
error sending request for url (https://edgeconfigapi.iot.maxmref.apps.68f21b70fe8d8ce89eab77bb.am1.techzone.ibm.com/api/installations/c4cdcd1-2af6-4180-8ac4-6c3ffc398392/)
Caused by:
0: client error (Connect)
1: error:1416f886:SSL routines:tls_process_server_certificate:certificate verify failed:ssl/statem/statem_clnt.c:1915: (unable to get local issuer certificate)
2: error:1416f886:SSL routines:tls_process_server_certificate:certificate verify failed:ssl/statem/statem_clnt.c:1915:
tanmayparashar@Tanmays-MacBook-Pro-2 ~ %
```

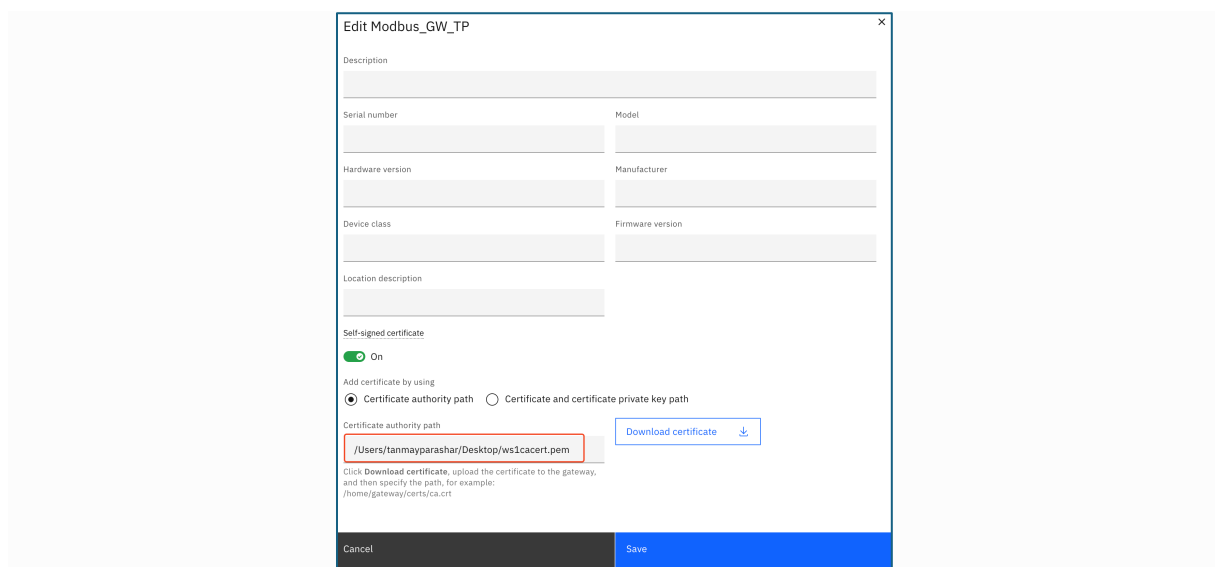
3. To fix the above error, go to Gateways you created in Monitor and click on Manage Gateway (three dots) and then click on **Edit**.



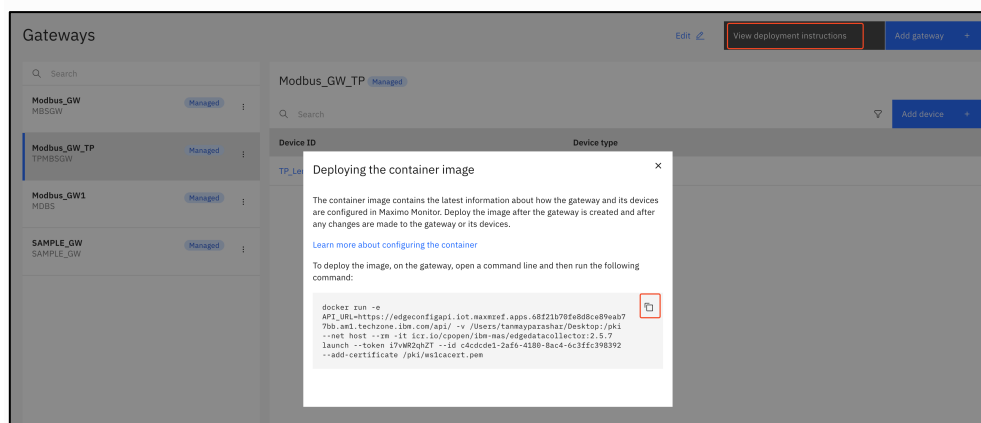
4. Click on **Edit**.



5. Enable **Self-signed certificate** and download the certificate from OCP (Workloads -> Secrets -> Project -> IoT -> Search public TLS -> Download ca.cert) and put the path of the certificate.



6. Click on **Save** and go to **Gateways** and copy the docker command by clicking on View deployment instructions.



- Open a terminal window (Mac/Linux) or Command window (Windows) where you want to run the Managed Gateway and then paste the docker command line from the clipboard. Click **enter** to execute it, and you should see something similar to the following.

```
tanmayparashar@Tanmayas-MacBook-Pro-2 ~ % docker run -e API_URL=https://edgeconfigapi.iot.maxmref.apps.68f21b70fe8d8ce89eab77bb.am1.techzone.ibm.com/api/ -v /Users/tanmayparashar/Desktop:/pk i
--net host --rm -it ior.io/cpopen/ibm-mas/edgedatacollector:2.5.7 launch --token 17WR2qh2T --id c4cdcd1-2afe-4188-8ac4-6c3ffc398392 --add-certificate /pki/wslcacert.pem
WARNING: The requested image's platform (linux/amd64) does not match the detected host platform (linux/arm64/v8) and no specific platform was requested
2025-10-26T18:38:13.883833Z INFO: edge-cli/src/lib.rs:108: Arguments: launch --token 17WR2qh2T --id c4cdcd1-2afe-4188-8ac4-6c3ffc398392 --add-certificate /pki/wslcacert.pem
2025-10-26T18:38:13.886393Z INFO: edge-cli/src/lib.rs:108: Version: 2.5.7
2025-10-26T18:38:13.886456Z INFO: edge-cli/src/lib.rs:109: Git sha1: cab6ca6f2bf101743db0f093cd83bb9763bc4668
2025-10-26T18:38:13.886495Z INFO: edge-cli/src/lib.rs:110: Branch: master
2025-10-26T18:38:13.886530Z INFO: edge-cli/src/lib.rs:111: Build number: 748
2025-10-26T18:38:13.886567Z INFO: edge-cli/src/lib.rs:112: Build ID: 41027338
2025-10-26T18:38:14.014018Z INFO: /omniot/edge/edge-rest/src/lib.rs:277: GET https://edgeconfigapi.iot.maxmref.apps.68f21b70fe8d8ce89eab77bb.am1.techzone.ibm.com/api/installations/c4cdcd
e1-2afe-4188-8ac4-6c3ffc398392/
2025-10-26T18:38:15.317381Z INFO: /omniot/edge/edge-rest/src/lib.rs:306: POST https://edgeconfigapi.iot.maxmref.apps.68f21b70fe8d8ce89eab77bb.am1.techzone.ibm.com/api/encrypted-profiles/
71d-d89d4184-2a49-4b10-b21b-fe3da28bddd8&id=af5fcebf-d81e-498c-a121-1963292f7ff&id=708367f8-f6a9-498b-9588-ccf7e4b9541&id=db6f8809-977c-4709-80eb-8bf7b131ee8&id=d3ab10d3-77fd-4b44-9e78-e
55ab69370f
2025-10-26T18:38:17.893537Z INFO: /omniot/edge/edge-rest/src/lib.rs:306: POST https://edgeconfigapi.iot.maxmref.apps.68f21b70fe8d8ce89eab77bb.am1.techzone.ibm.com/api/encrypted-profiles/
71d-8c4338e0-2937-6642-86da-099de7ab814&id=17a953b-7e8a-44b8-a6f3-19a5f16cccf3&id=f8011449-387b-486c-bf20-ce019d95976&id=f8ba2951-2f8d-4da9-be95-d21972dd1e1&id=94b8dac5-7d22-4396-8c2e-e
e095277c25a
2025-10-26T18:38:22.388366Z INFO: /omniot/edge/edge-rest/src/lib.rs:306: POST https://edgeconfigapi.iot.maxmref.apps.68f21b70fe8d8ce89eab77bb.am1.techzone.ibm.com/api/encrypted-profiles/
71d-08311173-ed6d-4172-b9f1-bb1563f64ca&id=511b590c-8276-498b-bf2f-0e67ca8f22f4&id=c37254a-7027-4445-ef2e-84b5209bcbdc&id=e1872f91-74c9-4952-b15d-81cd1f2ee49
2025-10-26T18:38:23.731137Z INFO: /omniot/edge/edge-rest/src/lib.rs:277: GET https://edgeconfigapi.iot.maxmref.apps.68f21b70fe8d8ce89eab77bb.am1.techzone.ibm.com/api/units/
2025-10-26T18:38:24.274589Z INFO: /omniot/edge/edge-rest/src/lib.rs:277: GET https://edgeconfigapi.iot.maxmref.apps.68f21b70fe8d8ce89eab77bb.am1.techzone.ibm.com/api/devices/923f1655-aa0
7-4d8c-8220-628ab3ac9b8f
2025-10-26T18:38:24.741937Z INFO: /omniot/edge/edge-rest/src/lib.rs:277: GET https://edgeconfigapi.iot.maxmref.apps.68f21b70fe8d8ce89eab77bb.am1.techzone.ibm.com/api/manufacturers/f523e2
c4-6841-4585-a55d-2ae3dca8e99a/
2025-10-26T18:38:25.527141Z INFO: edge-gateway/src/lib.rs:139: Connecting to modbus service at address '192.168.1.3:18502'
2025-10-26T18:38:25.530532Z INFO: edge-gateway/src/lib.rs:152: Connected to modbus service at address '192.168.1.3:18502'
2025-10-26T18:38:25.535890Z INFO: edge-publisher/src/publisher.rs:54: Connecting to mqtt service at address 'ssl://maxmref.messaging.iot.maxmref.apps.68f21b70fe8d8ce89eab77bb.am1.techzone.
ibm.com:443'
2025-10-26T18:38:27.089117Z INFO: edge-publisher/src/publisher.rs:67: Connected to mqtt service at address 'ssl://maxmref.messaging.iot.maxmref.apps.68f21b70fe8d8ce89eab77bb.am1.techzone.i
bm.com:443'
```

- Verify the selected Lenze data in Monitor. Click to open the **XX_Lenze_i550_01** device.

Device types

DVC_DB_TP

Analysis off

:

Lenze_DT_TP

Analysis off

:

LENZE_VFD

Analysis off

:

TP_DVC_TYPE

Analysis off

:

Lenze_DT_TP

Search

Device

Device name (Alias)

Description

TP_Lenze_i550_01

- Navigate to **Recent event** and wait for a minute until the first message is coming through.

Device type /

TP_Lenze_i550_01

Device

Analysis off

Manage device type

Data

Overview

Meters mapping

Metrics

Calculated metrics

Recent event

Dashboards

Dimensions

Event	Value	Format	Last received
status	["timestamp":"2025-10-26T18:48:13.369227Z","ControlCardTemperature_degreeCelsius":23.2000... View payload	json	a few seconds ago
status	["timestamp":"2025-10-26T18:48:12.344413Z","ControlCardTemperature_degreeCelsius":23.2000... View payload TP_Lenze_i550_01	json	a few seconds ago
status	["timestamp":"2025-10-26T18:48:11.315146Z","ControlCardTemperature_degreeCelsius":23.2000... View payload	json	a few seconds ago
status	["timestamp":"2025-10-26T18:48:10.289967Z","ControlCardTemperature_degreeCelsius":23.2000... View payload	json	a few seconds ago
status	["timestamp":"2025-10-26T18:48:09.261415Z","ControlCardTemperature_degreeCelsius":23.2000... View payload	json	a few seconds ago
status	["timestamp":"2025-10-26T18:48:08.233504Z","ControlCardTemperature_degreeCelsius":23.2000... View payload	json	a few seconds ago
status	["timestamp":"2025-10-26T18:48:07.214045Z","ControlCardTemperature_degreeCelsius":23.2000... View payload	json	a few seconds ago
status	["timestamp":"2025-10-26T18:48:06.190359Z","ControlCardTemperature_degreeCelsius":23.2000... View payload	json	a few seconds ago



Note- If the recent event is not showing any payload. Then go to the terminal where you run the docker command and copy the MQTT service append https before the URL and launch in the browser like:

```
2025-10-26T18:38:25.535800Z: INFO: edge-publisher/src/publisher.rs:54: Connecting to mqtt service at address 'ssl://maxmref.messaging.iot.maxmref.apps.68f21b70fe8d8ce89eab77bb.am1.techzone.ibm.com:443'
2025-10-26T18:38:27.009117Z: INFO: edge-publisher/src/publisher.rs:67: Connected to mqtt service at address 'ssl://maxmref.messaging.iot.maxmref.apps.68f21b70fe8d8ce89eab77bb.am1.techzone.ibm.com:443'
```

<https://maxmref.messaging.iot.maxmref.apps.68f21b70fe8d8ce89eab77bb.am1.techzone.ibm.com:443>

and accept the certificate. Refresh the device you will see the payload.

10. Click on **View payload** and see the data points being send to the Event name status.

Event payload window showing event details and a JSON payload.

Event name: status
Time received: Oct 27, 2025 15:33:38

```
1 {
2   "timestamp": "2025-10-27T10:03:38.314551Z",
3   "ControlCardTemperature_degreeCelsius": 21.400000000000002,
4   "DCBusVoltage_volt": 395,
5   "FrequencyCommand_hertz": 36.65
6 }
```

11. To see the data in the metrics. Go to Data -> Control Card Temperature -> Data Table.

Device type / TP_Lenze_i550_01

ControlCardTemperature_degreeCelsius (Metric) Last updated: 27/10/2025 00:43:28

Unit: degree Celsius

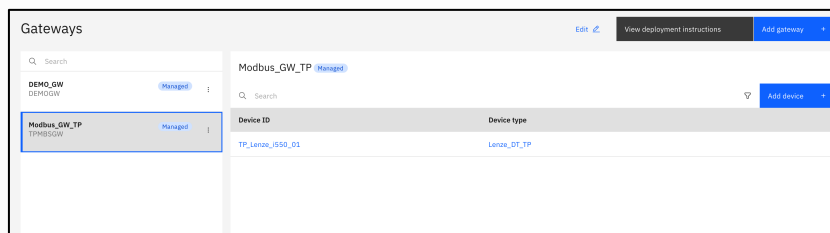
Trend Data table Dependencies

Timestamp	Device ID	ControlCardTemperature_degreeCelsius
27/10/2025 00:43:28	TP_Lenze_i550_01	23.8
27/10/2025 00:43:27	TP_Lenze_i550_01	23.8
27/10/2025 00:43:26	TP_Lenze_i550_01	23.8
27/10/2025 00:43:25	TP_Lenze_i550_01	23.8
27/10/2025 00:43:24	TP_Lenze_i550_01	23.8
27/10/2025 00:43:23	TP_Lenze_i550_01	23.8
27/10/2025 00:43:22	TP_Lenze_i550_01	23.8
27/10/2025 00:43:21	TP_Lenze_i550_01	23.8
27/10/2025 00:43:20	TP_Lenze_i550_01	23.8
27/10/2025 00:43:19	TP_Lenze_i550_01	23.8

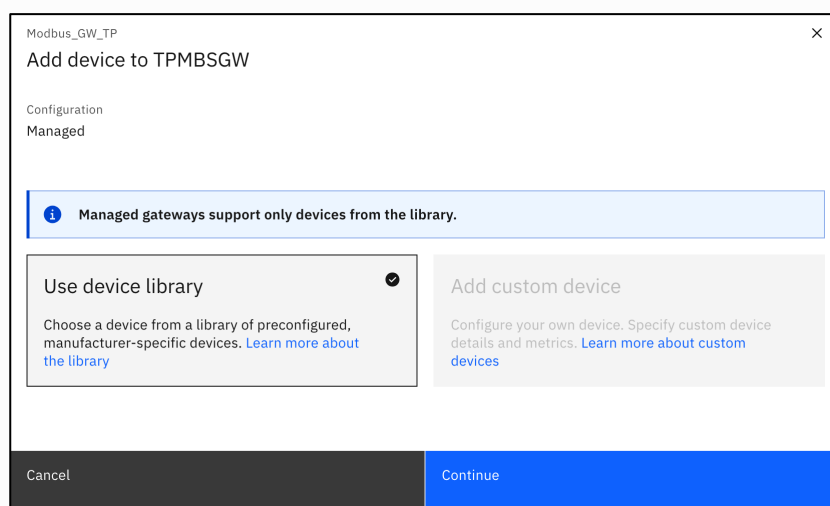
Exercise 5. Add another Industrial Device

In this Exercise you will learn how to add a second industrial device to the Managed Gateway. This time you will add it based on your Device Type rather than from the Managed Gateway.

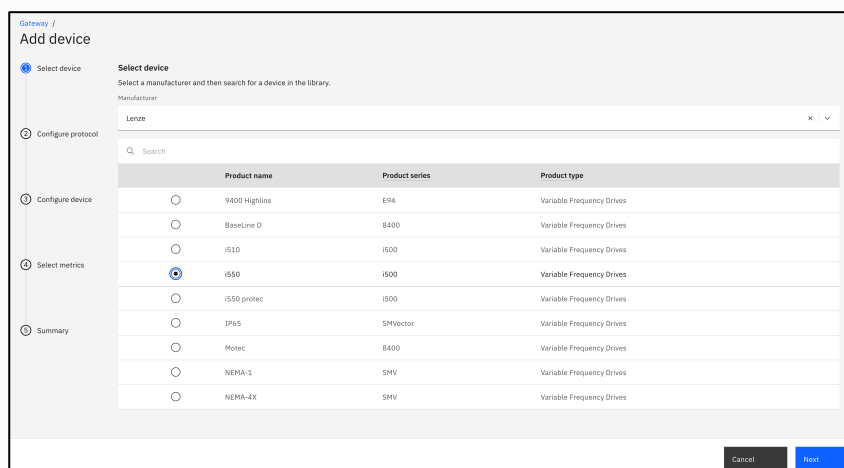
1. Open the Gateway in Monitor that you have created.



2. Click on **Add device**.



3. Select Manufacturer: **Lenze** and product: **i550** and click on Next.



4. Enter the Device ID and Device Type:

- Device ID: **XX_Lenze_i550_02**
- Device type: **Lenze_DT1_XX**

Gateway / Add device

Select device

Configure protocol

Configure device

Select metrics

Summary

Configure device

Device ID: TP_Lenze_i550_02

Device type: Lenze_DT1_TP

Product type: Variable Frequency Drives

Cancel Previous Next

5. Select these three metrics and click on Next.

Gateway / Add device

Select device

Configure protocol

Configure device

Select metrics

Summary

Select metrics

Select metrics for the device and configure when data is sent. Selected metrics are added to the device as row metrics. All of the metrics in the table are added to the selected device type. [Learn more about metrics](#)

Data frequency in milliseconds: 1000

Change threshold (%): On every change

3 items selected

Metric	Unit	Data frequency (ms)	Change threshold (%)
☐ Control Card Temperature	degree Celsius	1000	On every change
☐ DC Bus Voltage	volt	1000	On every change
☐ Frequency Command	hertz	1000	On every change
☒ Motor Current	ampere	1000	On every change
☒ Motor Torque	percent	1000	On every change
☒ Motor Voltage	volt	1000	On every change
☐ Output Frequency	hertz	1000	On every change
☐ Torque Command	percent	1000	On every change
☐ Supply Voltage	volt	1000	On every change

Cancel Previous Next

6. Review the Summary and click on Save.

Gateway / Add device

Select device

Configure protocol

Configure device

Select metrics

Summary

Summary

You're almost done! Here's a summary of the device configuration details for your confirmation.

Device Details

Manufacturer: Lenze

Product name: i550

Gateway ID: Modbus_GW_TP

Gateway type: TPMBSGW

Device ID: TP_Lenze_i550_02

Device type: Lenze_DT1_TP

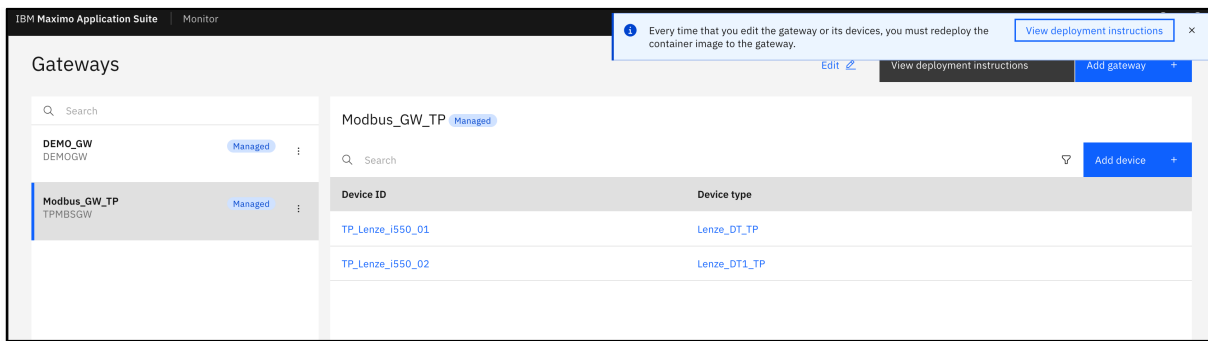
Protocol Details

Protocol: modbus-tcp

IP/Port/ServerId: 192.168.1.5:20502/1

Cancel Previous Save

7. Both the devices will be reflecting under the gateway.



Exercise 6. Deploy and view data in dashboard

In this Exercise you will learn how to:

- Stop and re-deploy the Managed Gateway
- View the incoming data in Monitor dashboards

1. Go to the Terminal or Command Prompt where the current Managed Gateway docker container is running.

[illegible]

2. Press **control-p** and then **control-q** to get back to the prompt.

3. Then run the command **docker ps** to view the running docker containers.

NAME	IMAGE	COMMAND	CREATED	STATUS	PORTS
156c9f2c1cb8	icr.io/cpopen/lbm-mas-edgeatacollector:2.5.7	"edge launch --token."	17 minutes ago	Up 17 minutes	
a3bfa4f3d312	elktron/modbus_simulator	"./entrypoint.sh"	59 minutes ago	Up 49 minutes (healthy)	0.0.0.0:10562->10562/tcp, [::]:10562->10562/tcp, 1880/tcp, 0.0.0.0:20562->20562/tcp, [::]:20562->20562/tcp
1c112b8232	4559/simulator	"./entrypoint.sh -gl"	2 hours ago	Up 2 hours	
8780e43f3f6	k8s_traefik-traefik-c98fd67b-vqxx_kube-system_c825d7d9-3eac-4563-8788-548f6b127a5_2	"local-path-provisioner"	2 hours ago	Up 2 hours	
31c87812fde6	b417c4f62d40	"metrics-server -m"	2 hours ago	Up 2 hours	
2e4c4597d2c1	bcd1e8e93841	"k8s_pod_local-path-provisioner-774c6d66dc-vjcw_kube-system_85dc8b26-76ea-4921-839f-486eb2cc571f_2"	2 hours ago	Up 2 hours	
b2828e755b8d	qos_metrics-server_metrics-server-70ffdc44-vyjm_kube-system_d4d0f71b-b679-4e6f-91dc-6a84643f8a3_2	"coredns -conf /etc/"	2 hours ago	Up 2 hours	
67336d26ad7b	a399cf1bd9d57	"k8s_coresdns.coredns-64fd464794-k6gvc_kube-system_582f9862-991a-40cf-b88e-886ef19bc27d_2"	2 hours ago	Up 2 hours	
ee227e649c58	81a2b0f7354a	"k8s_lb-top-443_svc1b-traefik-6c7b99f1-rrffc_kube-system_94a58bb3-f921-43ba-ab01-493b9778c6bf_2"	2 hours ago	Up 2 hours	
81a2b0f7354a	81a2b0f7354a	"./pause"	2 hours ago	Up 2 hours	
22c8a0779bc1	a399cf1bd9d57	"k8s_pod_local-path-provisioner-774c6d66dc-vjcw_kube-system_85dc8b26-76ea-4921-839f-486eb2cc571f_2"	2 hours ago	Up 2 hours	
848ee8199e8	81a2b0f7354a	"./pause"	2 hours ago	Up 2 hours	
12b5977ae843	81a2b0f7354a	"k8s_lb-top-443_svc1b-traefik-6c7b99f1-rrffc_kube-system_94a58bb3-f921-43ba-ab01-493b9778c6bf_2"	2 hours ago	Up 2 hours	
cad08aba4696	81a2b0f7354a	"./pause"	2 hours ago	Up 2 hours	

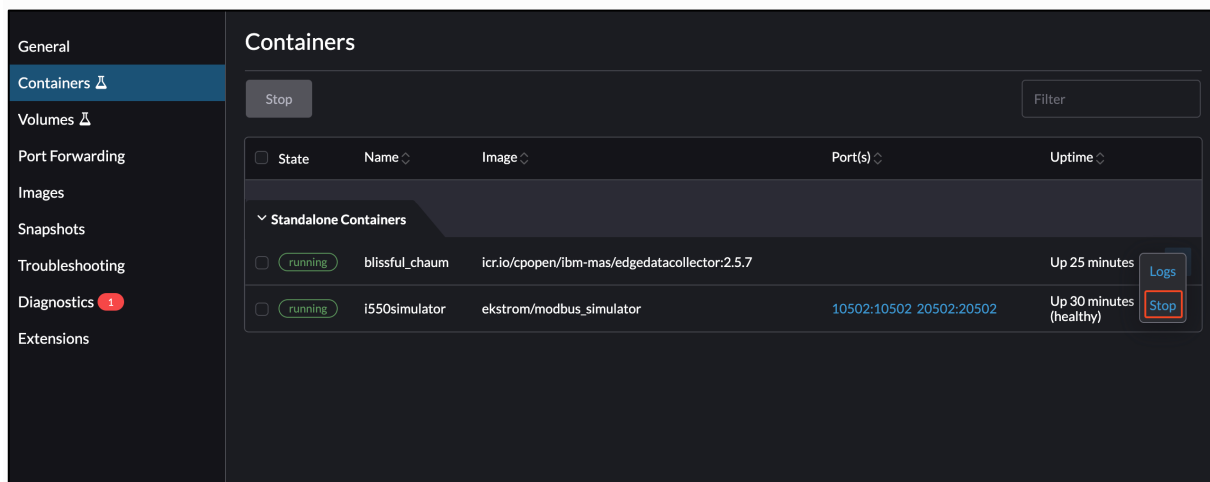
4. Copy the Container ID from the previous command i.e. 156c9fc21cb8.

And then run this command **docker kill <container id>**

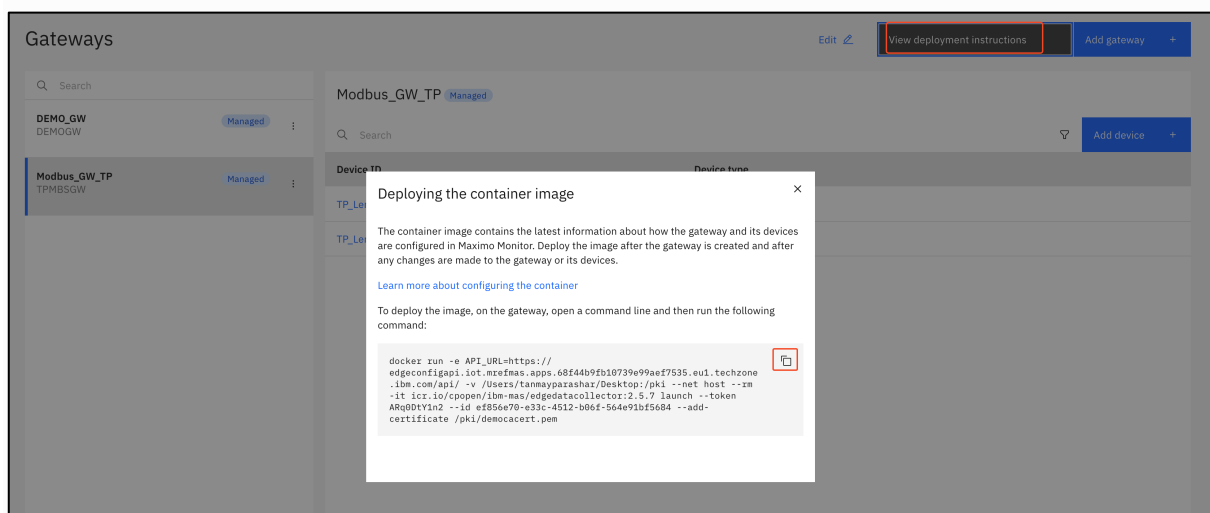
In this exercise the command would be: **docker kill 156c9fc21cb8**

```
tanmayparashar@Tanmays-MacBook-Pro-2 ~ % docker kill 156c9fc21cb8
156c9fc21cb8
```

OR You can directly kill the container from Rancher Desktop Application by clicking on Stop.



5. Navigate back to your Managed Gateway in Monitor and press the **View deployment instructions**. Click on the docker command to copy it to the clipboard.



6. Get back to the terminal and then paste the docker command line from the clipboard. Click enter to execute it, and you should see something similar to the following.

```
tanmayparashar@Tanmays-MacBook-Pro-2: ~ % docker run -e API_URL=https://edgeconfigapi.iot.mrefmas.apps.68f44b9fb10739e99aef7535.eu1.techzone.ibm.com/api/ -v /Users/tanmayparashar/Desktop/pk
1 --net host --rm -it ior.io/cpopen/ibm-mas/edgedatacollector:2.5.7 launch --token ARQ0DYIn2 --id ef85de78-e33c-4512-b86f-564e91bf5684 --add-certificate /pk1/democacert.pem
WARNINQ: The requested image's platform (linux/amd64) does not match the detected host platform (linux/arm64/v8) and no specific platform was requested
2025-10-27T18:38:14.734948Z INFO: edge-cli/src/lib.rs:186: Arguments: launch --token ARQ0DYIn2 --id ef85de78-e33c-4512-b86f-564e91bf5684 --add-certificate /pk1/democacert.pem
2025-10-27T18:38:14.737955Z INFO: edge-cli/src/lib.rs:108: Version: 2.5.7
2025-10-27T18:38:14.738028Z INFO: edge-cli/src/lib.rs:109: Git sha: cab0ca6f2bf01743db0f93cd83ab9763bc4668
2025-10-27T18:38:14.738092Z INFO: edge-cli/src/lib.rs:110: Branch: master
2025-10-27T18:38:14.738094Z INFO: edge-cli/src/lib.rs:111: Build number: 748
2025-10-27T18:38:14.738129Z INFO: edge-cli/src/lib.rs:112: Build ID: 41827330
2025-10-27T18:38:14.869495Z INFO: /omniot/edge/edge-rest/src/lib.rs:277: GET https://edgeconfigapi.iot.mrefmas.apps.68f44b9fb10739e99aef7535.eu1.techzone.ibm.com/api/installations/ef85de
78-e33c-4512-b86f-564e91bf5684/
2025-10-27T18:38:15.978211Z INFO: /omniot/edge/edge-rest/src/lib.rs:386: POST https://edgeconfigapi.iot.mrefmas.apps.68f44b9fb10739e99aef7535.eu1.techzone.ibm.com/api/encrypted-profiles/
71d56c37254a-752f-445b-af2e-84b32d9cbbcd&id=08904184-2a49-4b10-b21b-fe3a28bddd&id=94b0dac3-7d22-439b-8c2e-eeb9527cc25a&id=afc5fceb-d81e-498c-a121-1963292f77f7&id=0631f173-aded-4172-b9f1-b
1553f94c4c
2025-10-27T18:38:16.986131Z INFO: /omniot/edge/edge-rest/src/lib.rs:386: POST https://edgeconfigapi.iot.mrefmas.apps.68f44b9fb10739e99aef7535.eu1.techzone.ibm.com/api/encrypted-profiles/
71d56c37254a-752f-445b-af2e-84b32d9cbbcd&id=08904184-2a49-4b10-b21b-fe3a28bddd&id=94b0dac3-7d22-439b-8c2e-eeb9527cc25a&id=afc5fceb-d81e-498c-a121-1963292f77f7&id=0631f173-aded-4172-b9f1-b
1553f94c4c
2025-10-27T18:38:19.921568Z INFO: /omniot/edge/edge-rest/src/lib.rs:277: GET https://edgeconfigapi.iot.mrefmas.apps.68f44b9fb10739e99aef7535.eu1.techzone.ibm.com/api/units/
2025-10-27T18:38:19.281578Z INFO: /omniot/edge/edge-rest/src/lib.rs:277: GET https://edgeconfigapi.iot.mrefmas.apps.68f44b9fb10739e99aef7535.eu1.techzone.ibm.com/api/devices/923f1655-aa9
7-4d8c-822b-52ab3ac3b985/
2025-10-27T18:38:19.629908Z INFO: /omniot/edge/edge-rest/src/lib.rs:277: GET https://edgeconfigapi.iot.mrefmas.apps.68f44b9fb10739e99aef7535.eu1.techzone.ibm.com/api/manufacturers/f523e2
04-4841-4585-e355-2ac3dc98e99a/
2025-10-27T18:38:19.332888Z INFO: edge-gateway/src/lib.rs:139: Connecting to modbus service at address '127.0.0.1:18502'
2025-10-27T18:38:19.334331Z INFO: edge-gateway/src/lib.rs:152: Connected to modbus service at address '127.0.0.1:18502'
2025-10-27T18:38:19.337974Z INFO: edge-gateway/src/lib.rs:139: Connecting to modbus service at address '127.0.0.1:18502'
2025-10-27T18:38:19.338383Z INFO: edge-gateway/src/lib.rs:152: Connected to modbus service at address '127.0.0.1:18502'
2025-10-27T18:38:19.342028Z INFO: edge-publisher/src/publisher.rs:54: Connecting to mqtt service at address 'ssl://mrefmas.messaging.iot.mrefmas.apps.68f44b9fb10739e99aef7535.eu1.techzone.
ibm.com:443'
2025-10-27T18:38:20.795848Z INFO: edge-publisher/src/publisher.rs:67: Connected to mqtt service at address 'ssl://mrefmas.messaging.iot.mrefmas.apps.68f44b9fb10739e99aef7535.eu1.techzone.i
bm.com:443'
```



NOTE: In this step, the IP address is set to localhost. This situation occurs when the network does not have a fixed IP address. If you notice that your network configuration or IP address (as seen using the **ifconfig en0** command) changes frequently, switch to your local IP address (commonly **127.0.0.1**, though it may vary depending on your system). Then, open the device you have created in your setup and update its IP address accordingly.

7. Navigate to **Recent event** for the second device and wait for a minute until the first message is coming through.

Gateways / TP_Lenze_i550_02				
Device				
Analysis off Manage device type				
Data	Overview	Meters mapping	Calculated metrics	Recent event
Dashboards Dimensions				
Event	Value	Format	Last received	
status	["timestamp":"2025-10-27T11:17:19.404207Z","MotorCurrent_ampere":9.4,"MotorTorque_percent... View payload	json	a few seconds ago	
status	["timestamp":"2025-10-27T11:17:18.372952Z","MotorCurrent_ampere":9.4,"MotorTorque_percent... View payload	json	a few seconds ago	
status	["timestamp":"2025-10-27T11:17:17.360882Z","MotorCurrent_ampere":9.4,"MotorTorque_percent... View payload	json	a few seconds ago	
status	["timestamp":"2025-10-27T11:17:16.342997Z","MotorCurrent_ampere":9.4,"MotorTorque_percent... View payload	json	a few seconds ago	
status	["timestamp":"2025-10-27T11:17:15.322411Z","MotorCurrent_ampere":9.4,"MotorTorque_percent... View payload	json	a few seconds ago	
status	["timestamp":"2025-10-27T11:17:14.297568Z","MotorCurrent_ampere":9.5,"MotorTorque_percent... View payload	json	a few seconds ago	
status	["timestamp":"2025-10-27T11:17:13.270750Z","MotorCurrent_ampere":9.5,"MotorTorque_percent... View payload	json	a few seconds ago	
status	["timestamp":"2025-10-27T11:17:12.260398Z","MotorCurrent_ampere":9.5,"MotorTorque_percent... View payload	json	a few seconds ago	

8. Click on **View payload** and see the data points being send to the Event name status.

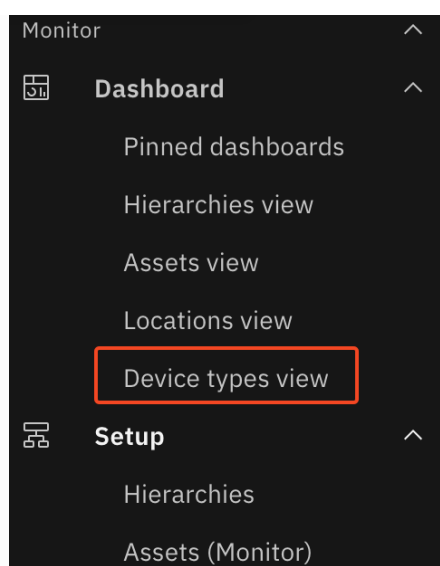


9. To see the data in the metrics. **Go to Data -> Motor Current -> Data Table.**

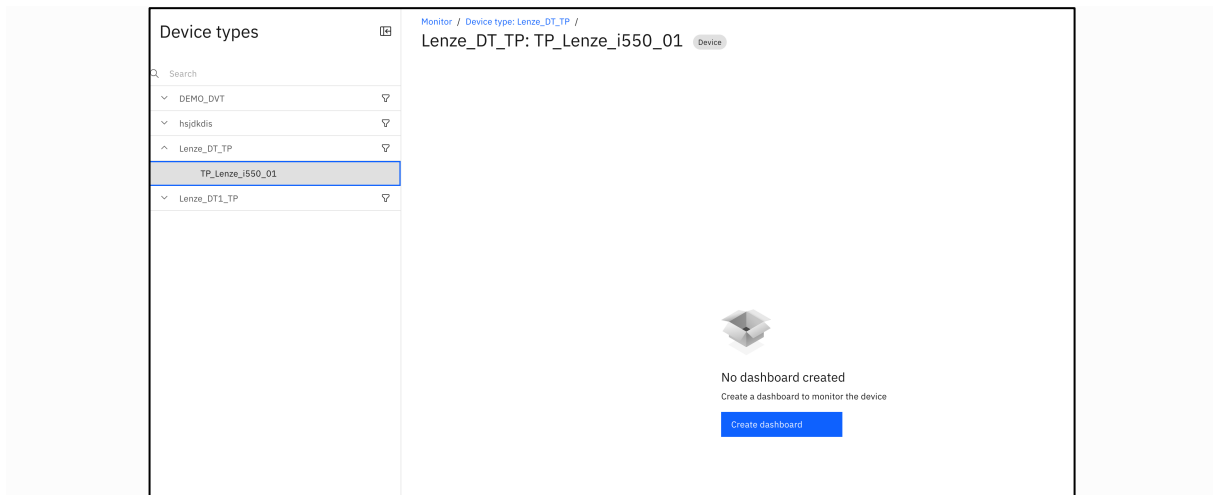
The screenshot shows the "Data" tab in the interface for device type "TP_Lenze_i550_02". The "MotorCurrent_ampere" metric is selected. The "Data table" view is active, showing a table with the following data:

Timestamp	Device ID	MotorCurrent_ampere
10/27/2025 16:51:01	TP_Lenze_i550_02	9.3
10/27/2025 16:51:00	TP_Lenze_i550_02	9.3
10/27/2025 16:50:59	TP_Lenze_i550_02	9.3
10/27/2025 16:50:58	TP_Lenze_i550_02	9.3
10/27/2025 16:50:57	TP_Lenze_i550_02	9.3
10/27/2025 16:50:56	TP_Lenze_i550_02	9.3
10/27/2025 16:50:55	TP_Lenze_i550_02	9.3
10/27/2025 16:50:54	TP_Lenze_i550_02	9.3
10/27/2025 16:50:53	TP_Lenze_i550_02	9.3

10. Navigate to **Dashboard -> Device Type View** in Monitor.

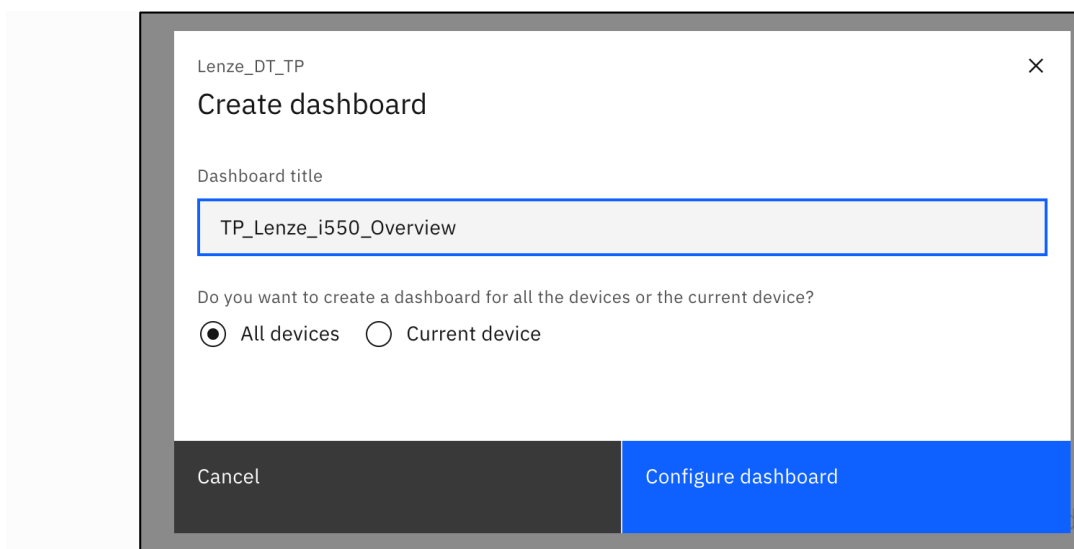


11. Select the **XX_Lenze_i550_01** device and then click on **Create Dashboard**



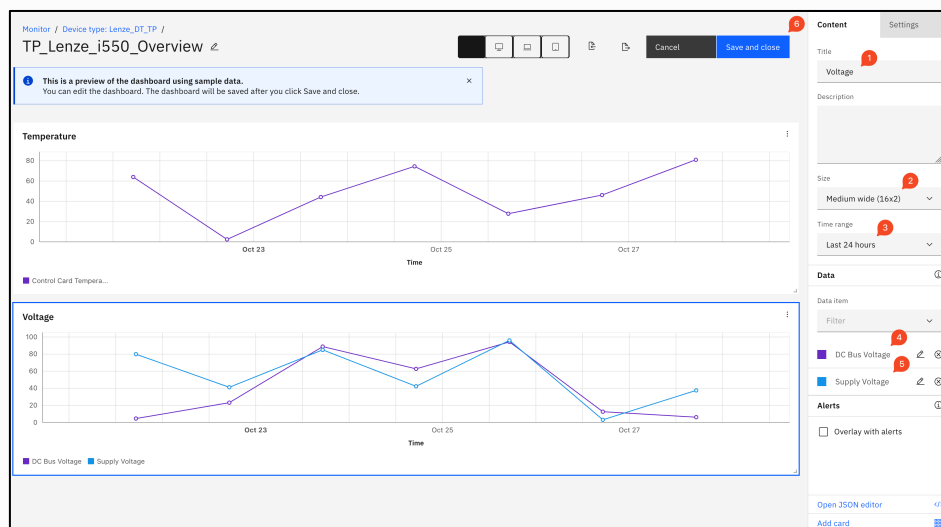
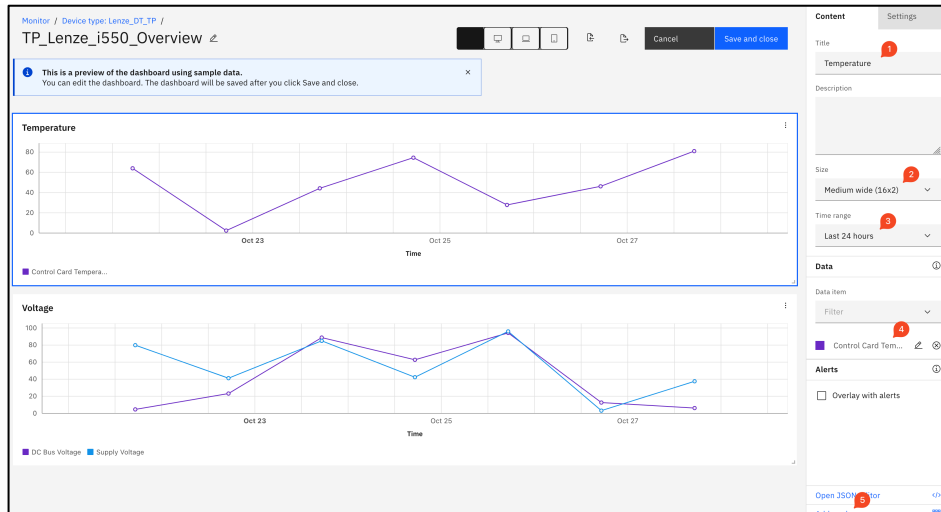
12. Give your new dashboard a name and press **Configure dashboard**.

Title: **XX_Lenze_i550_Overview**

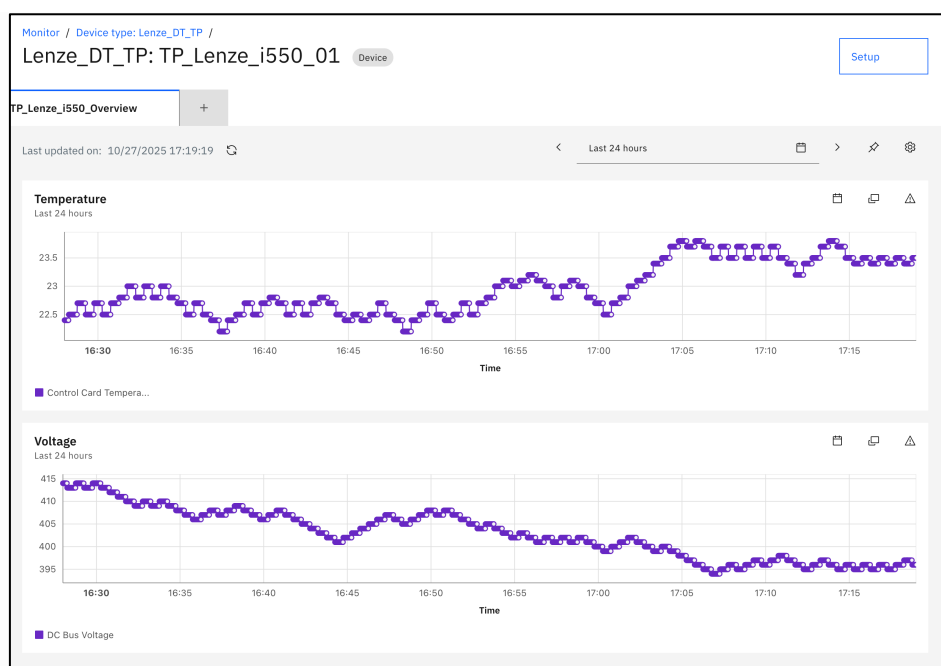


13. You will now see an empty dashboard configuration. Add a Time series line card by clicking on it.

- Locate the Control Card Temperature in the Data item dropdown and change the Time range to Last 24 hours.
- Give the card a title, like Temperature [°C] and drag the lower right corner to resize the card to full width.
- Once that is in place, then Add another Time series line card.
- Configure the second Time series line card according to the screen shot below and press Save and Close.



14. The dashboard for the first device will look like this.



15. Similarly you can configure for second device for different metrics.

